



PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale

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A programming plan is a template that highlights the goal of a common code pattern.

Motivation

Studies show plans can improve motivation [1], problem-solving skills [2]

But, they:

Have mostly been identified for introductory programming content

Are time-consuming and unclear to apply as an instructor

We propose:

Instructor-in-the-loop systems for interacting with LLMs can support this process.

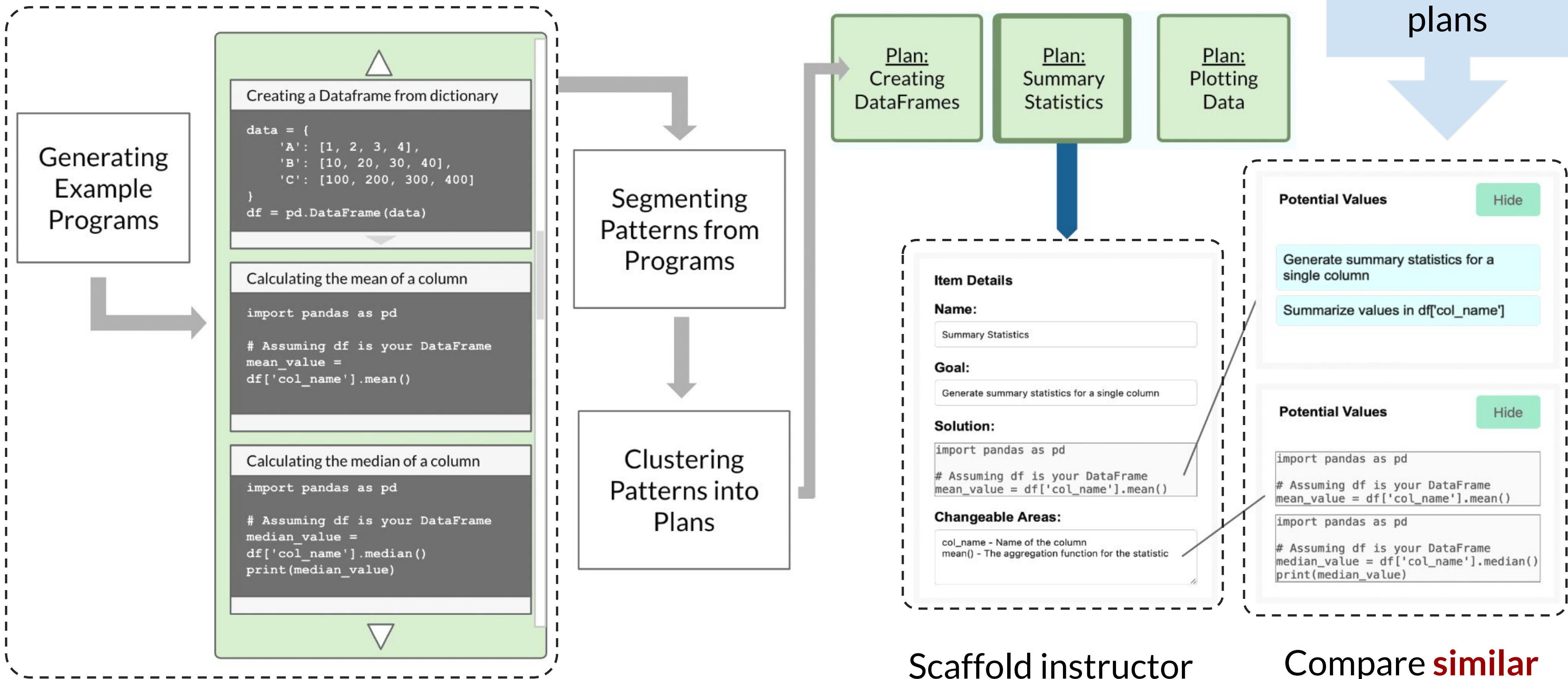
[1] K. Cunningham et al. "Avoiding the Turing Tarpit: Learning Conversational Programming by Starting from Code's Purpose."
[2] N. Weinman, A. Fox, and M. A. Hearst. 2021. Improving Instruction of Programming Patterns with Faded Parsons Problems. CHI '21.

PLAID enables using plan-based pedagogies in application-focused domains by presenting LLM-generated content and supporting interactions for its refinement.

Tedious for instructors to find plans from practice

Challenging to learn how to apply plan-based pedagogies

Difficult to identify the right level of abstraction in plans



Generate **many authentic programs** and enable **interactions** for quick exploration

Scaffold instructor experience by providing **structured components**

Compare **similar content** to find **high-level patterns**

Evaluation

Participants created **more plans** when using PLAID compared to the baseline condition.

Participants reported high scores on the **PSSUQ** usability survey.

Average task load for instructors was significantly **lower** with PLAID.

Takeaway

As part of instructor-in-the-loop systems, LLMs can automate repetitive work, allowing instructors to use their expertise to focus on refining content.